

1-Skin damage → release of PPS from the damaged cells → diffuse in the damaged area → lower the threshold of pain receptors → hyperalgesia. 2-Local axon reflexes: skin damage → afferent discharge of pain impulses → pass to CNS → pain sensation. Some impulses pass through branches of afferent fibers back to the area of skin damage (antidromic conduction) to end on : i-Skin arterioles → VD (Red Flare). ii-Cutaneous cells (mast cells) → release of PPS → lower the threshold of pain receptors in the area → hyperalgesia.	1-Cortical neurons for the area of secondary hyperalgesia are laterally inhibited by collateral neurons from area of primary hyperalgesia for the sake of sharp localization, hence the threshold of pain in area of secondary hyperalgesia is increased. 2-Convergence-facilitation theory: it explains why the response for pain from the area of secondary hyperalgesia is more intense than normal. -Afferents from area of primary hyperalgesia (through collaterals) converge on the same central neuron (SGR, Thalamus or CC) for the area of secondary hyperalgesia increasing their excitability
iii-The hyperexcitable pain receptors discharge at a higher rate and the feeling of pain becomes intense.	(facilitation) → intense pain response.

3. Physiology of Pain (II)

Control of pain sensation:

Pain pathway can be inhibited as evidenced by:

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- Stress analgesia: Soldiers wounded in the battle field may feel no pain until the battle is over.
- Rubbing of the skin around the injured area decrease the degree of pain sensation.
- Morphine is used as a powerful analgesic drug to reduce pain sensation.

GATE THEORY:

Mechanism:

Control of pain depends on the GATE THEORY FOR PAIN as follows: The relay between the first and second order neurons conducting pain impulses at the SGR involves a chain of several short ICN in between. The impulses should pass through several synapses or gates. These gates offer resistance to conduction of pain impulses and are under control from spinal and supraspinal pathways as follows:

Spinal And Supraspinal Pathways:

1) Spinal control of pain sensation:

The afferent nonmyelinated C fibers carrying impulses for delayed burning pain sensation activate the neurons of DRG (2nd order neurone through the release of a chemical transmitter (substance p).

Collaterals from large myelinated A β fibers that enter the dorsal column carrying tactile sensory impulses pass to end on the termination of the C fibers for pain and produce presynaptic inhibition $\rightarrow \downarrow$ the release of substance p $\rightarrow \downarrow$ pain transmission at the SGR $\rightarrow \downarrow$ pain sensation.

Clinical Significance:

- This explains why rubbing the skin around the injured area decreases

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pain sensation.

- Counter-irritation applied to the skin over a diseased deep organ relieves pain.
- High frequency electric vibrations applied to the surface of skin as in ACUPUNCTURE can block pain transmission (complete anaesthesia) for performing surgical operations.

2) Supraspinal control of pain

sensation (Endogenous Opiate

SYSTEM):

- Damaging stimulus → ascending pain impulses along the LST → act via collaterals on specific brain stem neurons to release ENDORPHINS AND ENKEPHALINS (opiate peptides) that will diffuse, bind and stimulate opiate receptors in 4 brain stem sites:
 1. Periventricular neurons.
 2. Paraventricular neurons,
 3. Substantia Nigra and
 4. Dorsal Raphe Nucleus (DRN).
- DRN when stimulated send descending impulses to the spinal cord along the dorsolateral funiculus (serotonergic fibers) to end either directly or through ICN on the SGR at the tip of the dorsal horn → release of serotonin → presynaptic inhibition of pain pathway → ↓ release of substance p → block the gate for pain sensation.

Other possible sites of release and action of endorphins and enkephalins:

- They are released from immune cells at site of injury → ↓ excitability of

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pain receptors.

- They are released at the tip of dorsal horn (SGR) from central branches of the DRG stimulated by collaterals of afferent pain fibers → presynaptic inhibition of afferent pain pathways at SGR → ↓ the release of substance p → ↓ pain sensation.

DEEP PAIN

- It is dull aching non localized pain.
- It is aroused by injury to deep structures as: skeletal muscles, ligaments, joints and periosteum of bones.
- It is conducted by C nonmyelinated fibers present in deep sensory motor nerves.
- It is associated by body reactions:
 1. Somatic: reflex immobilization.
 2. Autonomic: usually parasympathetic.
 3. Emotional: variable according to individual.
 4. Hyperalgesia over the diseased part.

Causes of deep pain

1. Bone injury: fracture, inflammation,...etc.
2. Joint disease: trauma, inflammation.
3. Muscle ischemia: decreased blood flow to muscles produces pain as in:
 - A. Intermittent Claudication (IC):
 - In this condition, the patient complains of severe pain in the muscles of the lower limb as he walks rapidly or runs, that forces him to stop and rest. In

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severe progressive cases pain occurs even during rest.

- Causes: narrowing of small arterioles supplying the muscles of the lower limb due to * Inflammation as in Burger's disease or
*atherosclerosis.
- Mechanism: Accumulation of pps and metabolites especially substance p, K^+ , and lactic acid during ischemic contraction → stimulate pain receptors in the muscle during activity. During rest, these metabolites are washed by increased blood flow → ↓ pain sensation.

B. Muscle spasm: as in tetany.

VISCERAL PAIN

- Most viscera contain only pain receptors (never contain proprioceptors, but may contain few thermoreceptors and touch).
- Certain viscera do not contain pain receptors and are insensitive to pain e.g. lung alveoli, liver parenchyma and visceral layers of serous membranes.
- Characters of visceral pain:
 - It is dull aching or colicky (hollow viscera).
 - It is diffuse and not localized.
 - It is usually referred to skin area supplied by the same posterior root (dermatome) as the diseased viscus.
- It is associated by body reactions:
 - Somatic: reflex rigidity (immobilization).
 - Autonomic: sympathetic or parasympathetic according to the degree of pain.

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- Emotional reactions: variable.
- Cutaneous hyperalgesia.

Causes of visceral pain

A. Ischemia: due to accumulation of metabolites and pps. This occurs in:

- Thrombosis of blood vessels supplying the viscus as in mesenteric artery or coronary thrombosis.
- Spasm of a hollow viscus as in ureteric or intestinal spasm → compression of blood vessels.
- Over distension of a hollow viscus → over stretching the wall → collapse of blood vessels e.g. over filling of the urinary bladder causes pain.

B. Chemical irritation:

- Perforated gastric ulcer → leakage of gastric HCl into peritoneal cavity → chemical irritation → pain.
- Pancreatitis → leakage of pancreatic proteolytic enzymes into peritoneal cavity → chemical irritation → pain.

Pathway of visceral pain

Pain impulses arising from FNE in viscera are conducted along C nonmyelinated fibers that run with:

1. The Sympathetic Nerves: These fibers carry the visceral pain impulses from most of the thoracic and abdominal viscera.

2. The Parasympathetic Nerves:

- Afferent visceral pain fibers from the pharynx, trachea, and upper oesophagus run with the glossopharyngeal and vagus nerves to

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enter the CNS directly at the level of the brain stem.

- Afferent visceral pain fibers from the pelvic organs : the distal portion of the GIT, the distal portion of the urinary system, the femal internal genital organs run with the sacral parasympatrhetic nerves.

3. The somatic Nerves: afferent pain fibers from the diaphragm and lower oesophagus run with the frenic nerve.

The cell bodies of all these fibers are present in the DRG and the central branches of these fibers enter the spinal cord along the lateral division of the posterior root → Lissauer's tract → follow the same pathway as burning pain sensation.

CLINICAL APPLICATIONS

1. Referred Pain (figure 18):

Visceral and deep pain is sometimes not felt at the site of origin but is referred to the skin area supplied by the same posterior root (dermatome) as the diseased viscus.

Examples of Referred Pain:

- i. Cardiac pain: is referred to the base of the neck, left arm, retrosternal or even epigastrium (T2-T5).
- ii. Gastric pain: is referred to skin area between the umbilicus to the xiphisternum (T7-T9).
- iii. Gall bladder pain: is referred to tip of the right shoulder (C3-C5) or to the hypochondrium (T7-T9).
- iv. Renal pain: is referred to the back, loin, inguinal region, testes and scrotum (T10-L1).

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- v. Appendicular pain: is referred to the umbilicus (T10) and right iliac fossa (L1).

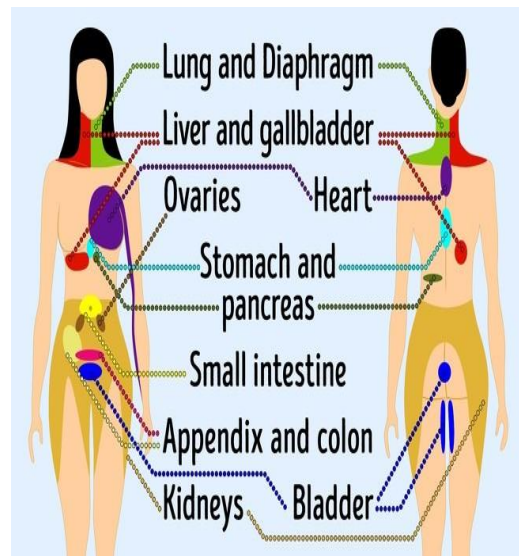
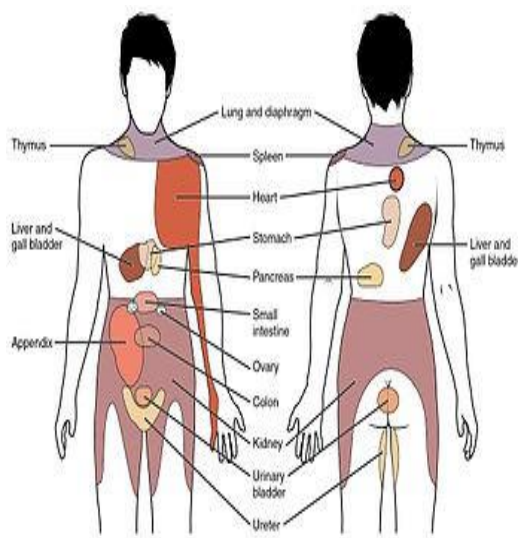


Figure 18: areas of referred pain

Mechanism (figure 19):

- Convergence-Projection Theory
- Convergence-Facilitation Theory

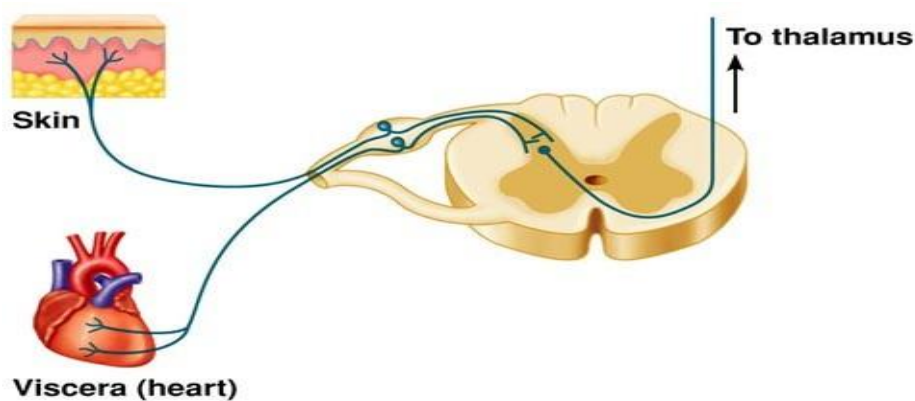


Figure 19: convergence projection theory

2. Itch and Tickle:

- Itching (pruritus) is not much of a problem for normal individuals, but

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severe itching that is difficult to treat occurs in diseases such as chronic renal failure, some forms of liver disease, atopic dermatitis, and HIV infection.

- Especially in areas where many naked endings of unmyelinated nerve fibers occur, itch spots can be identified on the skin by careful mapping.
- In addition, itch-specific fibers have been demonstrated in the ventrolateral spinothalamic tract.
- This and other evidence implicate the existence of an itch-specific path.
- Relatively mild stimulation, especially if produced by something that moves across the skin, produces itch and tickle.
- Scratching relieves itching because it activates large, fast-conducting afferents that gate transmission in the dorsal horn in a manner analogous to the inhibition of pain by stimulation of similar afferents. It is interesting that a tickling sensation is usually regarded as pleasurable, whereas itching is annoying and pain is unpleasant.
- Itching can be produced not only by repeated local mechanical stimulation of the skin but also by a variety of chemical agents. Histamine produces intense itching, and injuries cause its liberation in the skin.
- However, in most instances of itching, endogenous histamine does not appear to be the responsible agent; doses of histamine that are too small to produce itching still produce redness and swelling on injection into the skin, and severe itching frequently occurs without any visible change in the skin. The kinins cause severe itching.

3. Neuropathic pain:

Neuropathic pain may occur when nerve fibers are injured. Commonly, it is

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excruciating and a difficult condition to treat. It occurs in various forms in humans. For example, in causalgia, spontaneous burning pain occurs long after seemingly trivial injuries. The pain is often accompanied by hyperalgesia and allodynia. Reflex sympathetic dystrophy is often present as well. In this condition, the skin in the affected area is thin and shiny, and there is increased hair growth. Research in animals indicates that nerve injury leads to sprouting and eventual overgrowth of noradrenergic sympathetic nerve fibers into the dorsal root ganglia of the sensory nerves from the injured area. Sympathetic discharge then brings on pain. Thus, it appears that the periphery has been short-circuited and that the relevant altered fibers are being stimulated by norepinephrine at the dorsal root ganglion level. Alpha-adrenergic blockade produces relief of causalgia-type pain in humans, though for unknown reasons $\alpha 1$ -adrenergic blockers are more effective than $\alpha 2$ -adrenergic blocking agents. Treatment of painful sensory neuropathy is a major challenge and current therapies are often inadequate.

4. Muscle Pain:

- If a muscle contracts rhythmically in the presence of an adequate blood supply, pain does not usually result.
- However, if the blood supply to a muscle is occluded, contraction soon causes pain.
- The pain persists after the contraction until blood flow is reestablished.
- These observations are difficult to interpret except in terms of the release during contraction of a chemical agent (Lewis's "P factor") that causes pain when its local concentration is high enough. When the blood supply is restored, the material is washed out or metabolized. The identity of the P factor is not settled, but it could be K^+ . Clinically, the substernal pain that develops when the myocardium becomes ischemic during exertion (angina pectoris) is a classic example of the

accumulation of P factor in a muscle.

- Angina is relieved by rest because this decreases the myocardial O₂ requirement and permits the blood supply to remove the factor. Intermittent claudication, the pain produced in the leg muscles of persons with occlusive vascular disease, is another example.
- It characteristically comes on while the patient is walking and disappears on stopping.
- Visceral pain, like deep somatic pain, initiates reflex contraction of nearby skeletal muscle.
- This reflex spasm is usually in the abdominal wall and makes the abdominal wall rigid. It is most marked when visceral inflammatory processes involve the peritoneum. However, it can occur without such involvement.
- The spasm protects the underlying inflamed structures from inadvertent trauma. Indeed, this reflex spasm is sometimes called “guarding.”

4. Stretch reflex (I)

STRETCH (Myotatic) REFLEX (figure 20)

- Stretch of any skeletal muscle produces reflex contraction of the stretched muscle to oppose the stretch (increased length).
- It is a reflex contraction of a muscle when it is passively stretched.
- Receptors: the muscle spindle
 - Pathway